

The Accidental Agilists: One Team's Journey from Waterfall to Agile

Mary Beth Snapp
The Ohio State University
Snapp.6@osu.edu

Diane Dagefoerde
The Ohio State University
Dagefoerde.2@osu.edu

Abstract

This case study outlines one team's shift from the traditional waterfall methodology to an agile approach for web and custom application development. The transformation occurred by accident over the course of a web advising system project at The Ohio State University. Using the five stages of grieving as a metaphor, we will describe how the team moved from denial that the waterfall approach was failing to acceptance that agile practices would be the best way to deliver the mission-critical application. In the end, the entire team re-envisioned itself, transformed its business practices, and has evolved into a significantly more agile organization.

1. Project Background

Roughly 130 academic advisors at The Ohio State University's Colleges of the Arts and Sciences (ASC) serve more than 16,000 actively enrolled undergraduate majors in any given quarter. Students meet with advisors for a variety of reasons, from reviewing their degree plans and selecting courses to preparing for graduation. While committed to providing excellent service, the advisors were feeling overwhelmed by the sheer volume of appointments (they ultimately learned in 2008 that over 39,000 appointments were scheduled the previous year!). They wanted to change the nature of their interactions with students from a reactive approach to a more proactive one.

In the winter of 2006, the ASC Executive Dean formed a research team composed of advisors, administrators, and technology representatives who went both on-line

and on-site to investigate how many of Ohio State's peer institutions were dealing with these same kinds of advising challenges. After site visits to Michigan State and to Penn State, the research team returned with a clear idea of what ASC advisors needed — an advising support system that would help them:

- handle record keeping and mundane tasks more efficiently,
- increase the communication among advisors across departments about each student,
- begin reaching out to students more proactively to prevent small issues from turning into time-consuming advising interventions,
- document the volume of appointments and the types of advising interactions for annual reporting back to the ASC deans, and
- assign students to specific advisors rather than following the existing first-come, first-served workflow.

In May 2006, the Executive Dean tasked two of its reporting units, the ASC Advising and Academic Services Office (AAS) and the ASC Technology Services Office (TSO) with developing a web-based application to meet the advisors' needs. This paper's authors were part of the TSO group.

In addition to delivering the functionality above, the new advising support system needed to be:

- completed within existing resources (staff, infrastructure, etc.),
- released by the start of autumn quarter with a fall-back position of December break (those were the only release windows available given the workload and workflow of the office).

Developing custom web applications with existing budgets and staff is a common practice in a higher education environment. Given its light project load at the time and its assessment of the project overall, TSO felt it was feasible with existing staff and could be completed by the autumn quarter deadline – with one condition.

Because the project was so highly visible, and the timeline was so tight, TSO felt it was critical to structure the project “the right way.” That is to say, TSO’s condition for the project was to use formal governance for quicker decisions and to structure the work using the waterfall approach for streamlined development.

In May 2006, TSO and AAS created a governance structure that was based on those of several large-scale ERP projects at Ohio State. The structure included four groups whose roles and responsibilities are outlined below.

Executive Steering Committee (three members: Executive Dean and leaders of AAS and TSO)

- Communicates project's goals and objectives to the university-wide audience
- Provides guidance to resolve issues that the core team cannot agree on
- Reviews recommendations from a university-wide perspective and provides feedback to the core team
- Provides final sign-off on the project design

Project Manager (AAS Administrator)

- Acts as the primary communicator of the project's goals and objectives to the front-line staff
- Monitors progress and provides status reports to administrators and other necessary parties
- Leads the core team

Core Team (eight members: at least one from each sub-unit of AAS; one from TSO)

- Defines, develops, and manages the project’s scope, deliverables and deadlines
- Forms Advisory Teams to gather requirements
- Forms feedback groups to help with design, testing and other phases of development
- Approves/prioritizes recommendations made by the Advisory Teams
- Communicates broadly across all groups

Implementation Team (three members from TSO)

- Executes the project based on the core team’s specifications
- Facilitates design/testing/feedback sessions

Advisory Teams (51 members total)

- Play a consulting role in the gathering of requirements by articulating advising tasks and workflows

In late May 2006 the core team met for the first time. During the meeting, TSO staff explained the traditional waterfall approach (consecutive phases of analysis, design, development, testing, deployment) and reviewed the commitments that would be required of various AAS staff along the way. TSO folks also explained how formal sign-offs would move the project from one phase to the next, and that changes would need to be approved by the executive steering committee. At the end of the meeting, the group decided on the project timeline below with much enthusiasm and good will.

Table 1: Original Project Timeline

Milestone	Week	Deadline
Approve scope document	Week 1	May 19
First core team meeting	Week 2	May 26
Analysis and design	Week 2 – 7	June 29
Approve user interface design	Week 7	June 29
Draft data model	Week 7	June 29
Define reports	Week 7	June 29
Develop application	Week 8 – 13	August 11
Write user documentation	Week 12	August 4
Train testers and test	Week 12 – 13	August 11
Finalize deployment plan	Week 13	August 11
Approve reports	Week 13	August 11
Fix bugs	Week 13 – 14	August 18
Training of all staff	Week 15 – 16	Sept. 1
Final signoff on application	Week 16	Sept. 1
Release		Sept. 5
Final Project Deadline		Sept. 5

The remainder of this paper is organized as follows:

- section two explains how the project's circumstances changed which forced the implementation team to abandon the waterfall approach in favor of "just winging it,"
- section three details TSO's reflections on the project and lessons learned
- section four outlines what TSO will do (and has done) differently in subsequent application development projects.

2. The Accidental Transition to Agile

The project timeline began to unravel during the analysis phase and seemed doomed as the end of June approached. Despite TSO's best efforts to rein the project back in with a stronger emphasis on planning, it became clear that more "radical" steps would need to be taken in order to deliver something of value by the September deadline.

When the project started deviating from the "perfect" plan, the team had to give up what it believed was the proper way to do software development. TSO felt it had, in fact, suffered a true "loss." In fact, TSO's transition away from the waterfall methodology could be likened to a journey through the stages commonly associated with the grieving process [1]:

- Denial
- Anger
- Bargaining
- Depression
- Acceptance

Denial: *"I can't believe this is happening!"* Early on in the project, red flags started popping up:

- The advising processes were undocumented, very complicated, and only a handful of the advisors knew them well.
- Some members of the core team did not feel empowered to make decisions.
- Key people were not able to give this project the time and the attention that it needed because of competing priorities.
- Some team members had never participated in a software development project and assumed that the tech team would be doing the work.
- TSO's web developer and key member of the implementation team left the university.

The response to these issues was to deny their seriousness. TSO posted for the new developer position, expecting to hire quickly. The core team divvied up the requirements-gathering among the team members, assuming that it would get done faster.

Anger: *"Why us? It's just not fair!"* By July, things really started going wrong, and frustration levels began to increase.

- After a lengthy interviewing process, TSO's first choice for a developer turned down the offer. Still no developer!
- There were huge gaps in understanding of some aspects of the advising business process.

How could TSO start designing a solution until the requirements documentation had been delivered?

Bargaining: *"Things will improve if only we try harder!"* In late July 2006, the TSO team was still clinging tightly to the belief — learned in college and through years of experience on traditional software development efforts — that the waterfall model was the "right" way to manage a project. In response to serious obstacles, TSO started forcing the traditional methodology onto the project with even more gusto. Everything will be better, if the core team could just finish that requirements documentation, if there were more meetings, if everyone worked more hours!

However, there were also blessings in disguise that helped to save the project:

1. An enthusiastic, highly motivated advisor unofficially moved to the implementation team and served as the representative of the customer. She was the "bridge" between the implementation team and the end users. Communication became very efficient, as she was available at all times to answer questions and translate requirements.
2. The TSO project manager (also part of the implementation team) had strong enough database skills to start designing a database and testing simple business logic with primitive web pages.

3. The director of TSO introduced the team to paper prototyping. The conference room became covered in large sheets of paper with sketches of web pages, and the requirements documents were filed away and forgotten.

Depression: *“This project is a failure!”* With the September 5th deadline looming, it seemed likely that nothing would be delivered by the start of autumn quarter. After all, there was still no user interface!

Of course, projects “slip,” but TSO was reluctant to admit its failure due to the high visibility of the project and TSO’s insistence on the waterfall approach.

Acceptance: *“It’s going to be OK.”* TSO’s new web developer started on August 1 and was immediately assigned to the implementation team. He quickly devised a professional and aesthetically pleasing interface for the web application. This was the “aha” moment for the team, the moment where delivering something of value by September seemed possible again.

With less than a month to go, the team focused its energies on the most critical features. The search interface was released early in September to a subset of the advising staff, so they could retrieve student academic information when scheduling appointments. In December, the scheduling interface was rolled out to the reception/scheduling team. The various features of the full advisor interface, including the ability to assign students to “rosters” and a tool to allow advisors to send emails to their advisees, were made available throughout January 2007. The Online Student Advising and Services Center system (OSASC) was live!

The system has been well-received. Advisors report that it is intuitive and easy to use. It has streamlined advising operations, and has enabled advisors to provide improved services to students. It was made available to the distributed department-level advisors in the summer of 2007. And the word is out: TSO continues to receive requests from advisors across Ohio State for access and training. It has been a great success!

3. Lessons Learned

With OSASC running smoothly, TSO had time to reflect on the project. The team wanted to determine why the end result worked so well when the waterfall process completely fell apart. The authors’ research brought them to the agile Manifesto and its four key values [2]:

1. Individuals and interactions over processes and tools
2. Working software over comprehensive documentation
3. Customer collaboration over contract negotiation
4. Responding to change over following a plan

In the authors’ opinion, it’s very likely that this project would not have delivered any business value by the deadline if the team had continued to force the waterfall approach. Fortunately for the project, circumstances surrounding it forced the team to operate in ways that reflect the agile values mentioned above.

- The 3-member development team consisted of highly collaborative “generalists.” All of them pitched in to get the job done.
- Advisors helped to create all of the paper prototypes and became very excited when they started to see concrete progress. Paper prototypes became our “ubiquitous language” and replaced the technical requirements documentation.
- Advisors thoroughly tested every interface and feature as it was being created. There was a continuous feedback loop between advisors and the developers as the interfaces were being programmed.
- The team remained flexible with features. As advisors’ understanding of the software evolved, their requirements changed. These changes were welcomed and implemented. In fact, changes to one problematic page were being made literally the day before deployment.

Another concept the authors discovered was that of time boxing. In waterfall, features tend to be fixed and costs and schedules/deadlines tend to change (expanding, typically) as the project progresses. In agile, what remains fixed are costs and schedules/deadlines. This forces features to change (contracting, typically) as cost and schedule limits are approached. In the case of OSASC, what was most fixed was the deadline which forced the team to stay focused on the highest priority features in order to deliver something of value by early September. All other feature requests were queued for future releases.

The authors agree that agile is a much better fit for higher education and non-profit environments than the waterfall approach. Budgets and deadlines in such organizations are almost always fixed and any application development work typically must be done within those constraints. Furthermore, the authors agree that agile's emphasis on emerging requirements, short iterations, increments of product functionality, collaboration and face-to-face communication, transparency, and responsiveness to change align very well with how administrative groups in higher education function.

4. What Will Be Different Next Time

In April 2008, TSO was contracted to develop specific modules of OSASC to the entire university as part of its PeopleSoft Student Information System project. Advisors across Ohio State will be able to record and share their notes about their interactions with the more than 55,000 students of the university. The delivery date is of course, the first week of September!

Because the project is significantly larger in scale and has some key differences in requirements from OSASC, a new application called Enterprise Notes will be developed.

The university project kicked off in May 2008 and is being developed using agile software development methods. TSO is applying agile principles and practices formally with Scrum as the project management framework and XP (Extreme Programming) for its engineering practices. Paper prototyping continues to play a central role in engaging end users and communicating designs among developers and stakeholders.

TSO also supports some 30 administrative web sites and each site re-design or new site request gets the agile/Scrum/XP treatment as well. TSO has even begun to apply some Scrum project management practices to its infrastructure and desktop support projects.

5. Conclusion

Organizational change is hard. As with any "loss," the journey from the well-entrenched waterfall model to agile software development will likely take you and your team through the stages described here [3]. Awareness and acceptance of these stages will ease the transition. As the authors learned through a highly visible, large scale project, the benefits of agile approaches to your organization are well worth the "grieving" involved in "losing" the traditional ways of producing software.

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7. References

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